

TRUSTED CARBON REDUCTION (TCR) CARBON STANDARD

Revision 0.4

PROJECT REVISIONS AND VERSIONS

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PREAMBLE

This document presents in a concise and structured manner the TCR Carbon Standard, together with its Methodology and Operational Framework, providing a clear reference framework to guide informed decision-making. At the same time, it highlights practical and measurable actions aimed at promoting the adoption of sustainable agricultural practices and the rigorous quantification of greenhouse gas reductions and removals.

The TCR Standard has been developed to support and facilitate voluntary CO₂ emission offsetting activities within the Voluntary Carbon Market (VCM). Unlike Mandatory Markets, which are regulated at state level in various regions of the world and focused on specific industrial sectors, Carborea does not operate in such regulated contexts, where each State independently defines rules, regulations and carbon quotas.

It is worth noting that, according to several studies and analyses, the quotas provided for in Mandatory Markets are not always effective in promoting the sustainability objectives set out in international agreements such as the Kyoto Protocol and the Paris Agreement. Some experts even point out that these instruments, in certain contexts, may have effects contrary to those desired, contributing to dynamics that are far from virtuous on the environmental front (Cames et al., 2016).

Carborea, on the other hand, operates exclusively in the voluntary context, leveraging the cardinal principle of Additionality: the demonstrable capacity of a project to generate CO₂ reductions or sequestrations that would not have occurred in the absence of the intervention. The TCR standard is designed to scientifically and rigorously quantify CO₂ absorptions and flows, on the basis of the methodologies and standards presented in this document.

The methodological approach of the TCR reflects a shared consensus among communities and stakeholders, according to which effective actions for climate protection can be taken at any scale, even by individuals or small groups. The mission of the TCR is to contribute concretely to environmental conservation through the restoration of soils, ecosystems and degraded areas, promoting interventions that:

- Reduce overall CO₂ emissions;
- Promote the sequestration of atmospheric carbon;
- Are aligned, while acting independently, with the objectives outlined by international climate agreements.

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1. INTRODUCTION

In an increasingly climate-conscious and environmentally responsible global context, Carborea stands out as an innovative company committed to developing blockchain-based solutions in the service of sustainability. Its mission is to leverage the transformative potential of blockchain to redefine the carbon credit ecosystem, ensuring transparency, traceability and integrity in environmental certification processes. Through this vision, Carborea intends to actively contribute to the fight against climate change, supporting the creation of reliable and accessible systems for the measurement, verification and exchange of CO₂ reductions and removals, in line with the needs of an economy increasingly oriented towards ecological transition.

1.1 CARBOREA: COMMITMENT TO SUSTAINABILITY

Carborea positions itself resolutely at the forefront of innovation in the field of blockchain applied to sustainability, placing environmental protection as a central and non-negotiable value. The team, composed of highly specialised professionals, integrates advanced blockchain technologies with a deep commitment to ecological integrity, developing tools that go beyond the simple facilitation of transactions. Carborea's objective is to act as an engine of positive and lasting change, promoting solutions that foster transparency, reliability and effectiveness in carbon credit-related processes. Convinced that the union between technological innovation and environmental management represents a strategic lever for the future, Carborea aims to overcome the limitations of traditional models, opening new perspectives for truly sustainable progress.

1.2 CONTEXT

The TCR Carbon Standard has been developed with the objective of providing structured and rigorous guidelines for the assessment, monitoring and issuance of carbon credits within the Voluntary Carbon Market (VCM). Particular attention is paid to Afforestation and Reforestation projects, recognised as key instruments for the removal of CO₂ from the atmosphere and for the restoration of degraded ecosystems, in line with the additionality and permanence principles inherent to the standard.

1.3 SCOPE AND EXTENT

The TCR methodology establishes in detail the requirements, formulas and operational procedures for the quantification, reduction and offsetting of greenhouse gas (GHG) emissions. It defines the criteria for the generation of certifiable carbon credits in the context of the Voluntary Carbon Market, ensuring compliance with the principles of transparency, additionality, environmental integrity, while at the same time promoting local social and economic benefits.

The TCR Carbon Standard applies to afforestation and reforestation (A/R) projects carried out throughout the national territory, with the objective of mitigating GHG emissions through the sequestration of atmospheric CO₂ through sustainable forestry practices. The methodology ensures that such projects not only generate measurable environmental benefits, but also ensure the long-term permanence of sequestered carbon, making a concrete contribution to climate resilience and ecosystem restoration.

1.4 THE CURRENT VOLUNTARY CARBON MARKET AND ITS CHALLENGES

The Voluntary Carbon Market (VCM) represents a strategic instrument in the global challenge against climate change, offering organisations and individuals the possibility of voluntarily offsetting their greenhouse gas emissions through the financing of projects with certified environmental benefits. However, the VCM faces structural challenges that limit its full potential:

Limited transparency: one of the main weaknesses of the market is represented by poor transparency and accountability, which can compromise the credibility and traceability of projects, raising doubts among stakeholders about the actual effectiveness of offsetting interventions.

Complexity and fragmentation: the presence of multiple standards, methodologies and registries has created a heterogeneous and often inaccessible ecosystem. This fragmentation can represent a barrier to entry for new actors and make it complex to compare different project approaches.

Challenges in verification and additionality: the independent verification of projects and the assurance of the principle of additionality – i.e. the demonstration that the environmental benefits would not have occurred in the absence of the project – represent critical elements. The heterogeneity of contexts and reference practices can make it difficult to apply these criteria in a uniform and consistent manner.

Through the TCR Standard and blockchain technology, Carborea aims to address these challenges by offering methodological rigour, verifiable transparency and an integrated operational structure, fostering a carbon market that is more accessible, reliable and effective.

1.5 HOW THE TCR STANDARD BENEFITS THE VOLUNTARY MARKET

The TCR standard represents a significant advancement in the landscape of the Voluntary Carbon Market (VCM), offering a concrete response to the main critical issues of the sector and introducing measurable operational and structural benefits:

Enhanced transparency and reliability: the Carborea platform, founded on blockchain technology, ensures complete traceability and immutability of data, allowing all stakeholders to monitor the entire lifecycle of carbon credits. This system guarantees maximum clarity and trust, helping to prevent fraud and strengthen the legitimacy of registered projects.

Process simplification: thanks to a standardised and scalable approach, the TCR methodology optimises and rationalises the stages of issuance, monitoring and retirement of carbon credits. This reduces operational barriers and promotes inclusive participation, even for small and medium-sized organisations.

Robust verification and additionality assessment: through the use of advanced digital technologies and dedicated algorithms, the Carborea system facilitates independent verification processes and enables an objective, repeatable and transparent assessment of the additionality criterion, an essential element for the environmental validity of the credits issued.

In the following sections, the main components of the TCR Methodology will be analysed in detail, highlighting how it redefines the standards of governance, integrity and innovation within the Voluntary Carbon Market. This path aims to accelerate the sustainable transition, leveraging the potential of blockchain to generate a positive and global environmental impact.

2. TERMS AND DEFINITIONS

TCR Carbon Standard

A structured set of guidelines and procedures developed by Carborea to ensure the governance, implementation and certification of carbon credit projects in the context of the Voluntary Carbon Market (VCM).

Afforestation

A practice aimed at the deliberate creation of forest areas on land that has not had forest cover in the recent past. It is carried out through the planting of trees and native vegetation to promote the sequestration of atmospheric CO₂ and combat climate change.

Reforestation

A process aimed at restoring forests on previously forested, degraded or deforested surfaces due to human activities or natural events. It aims to recover the original ecological functions and carbon storage capacity.

Voluntary Carbon Market (VCM)

An unregulated system in which organisations and individuals purchase carbon credits to voluntarily offset their greenhouse gas emissions, thereby contributing to environmental initiatives and sustainable projects.

Carbon Accounting

The process of quantifying the CO₂e removals or reductions achieved through project interventions, in particular in afforestation and reforestation (A/R) projects, according to standardised methodologies.

Carbon Credits

A unit representing one metric tonne of CO₂ equivalent removed, reduced or avoided through certified activities. Credits can be traded or retired to offset emissions produced elsewhere.

Reference Emissions (Baseline)

The emissions scenario expected in the absence of the project intervention. It serves as a benchmark for measuring the actual reduction or removal of greenhouse gases achieved by the project.

Additionality

A cardinal principle of the VCM that ensures that CO₂ reductions or removals are additional to those that would have occurred in the "business-as-usual" scenario. It demonstrates the actual environmental effectiveness of the project.

Losses (Leakage)

The involuntary displacement of emissions outside the project area due to the activities undertaken, with the potential to neutralise the environmental benefits. They must be identified, monitored and mitigated.

Permanence

The capacity of the project to maintain carbon sequestration or emission reductions over time. It addresses the risk that the stored carbon will subsequently be released into the atmosphere.

Permanence Buffer

A precautionary reserve of credits or sequestered carbon, intended to cover any future losses due to unforeseen events, thus ensuring the robustness and environmental continuity of the project.

Monitoring, Reporting and Verification (MRV)

A continuous system that includes the collection of project data, periodic reporting and independent verification to ensure compliance with standards and the reliability of declared results.

Blockchain Technology

A distributed digital ledger technology that enables the traceability, transparency and integrity of information, proving fundamental for the governance of carbon credits.

Smart Contracts

Self-executing contracts encoded on blockchain, which allow operations (such as the release of credits) to be automated based on predefined conditions, improving efficiency and trust.

Immutable Registry

A digital registry on blockchain in which each transaction, once recorded, cannot be modified or deleted, ensuring integrity, traceability and long-term transparency.

3. PRINCIPLES

The TCR Carbon Standard methodology offers a systematic and scientifically grounded approach for the assessment, measurement and verification of carbon sequestration in afforestation and reforestation (A/R) projects. The methodological system ensures that projects meet the eligibility criteria, demonstrate effective additionality, and accurately consider reference emissions, in full compliance with the standard's guiding principles: environmental integrity, socio-economic benefits, transparency and permanence.

The formulas and calculation criteria included in the methodology constitute the operational basis for the quantification, validation and verification of results achieved, ensuring the credibility and traceability of CO₂ sequestration in projects registered under the TCR.

3.1 ENVIRONMENTAL INTEGRITY

Environmental integrity represents a founding principle of the TCR Carbon Standard. In afforestation and reforestation (A/R) projects, this principle requires that climate action be closely linked to ecosystem protection and biodiversity conservation.

Projects must be designed and implemented in accordance with sustainable land management practices, ensuring that no negative impacts occur on natural habitats and that activities actively promote ecological resilience and the overall well-being of the environment. Environmental integrity, therefore, is not limited to the reduction of emissions, but extends to the quality and sustainability of the actions taken.

3.2 ASSOCIATED SOCIAL AND ECONOMIC BENEFITS

The principle of Social and Economic Benefits constitutes a key element of the TCR standard, affirming that afforestation and reforestation (A/R) projects must generate environmental value without neglecting the well-being of local communities.

In addition to contributing to the reduction of greenhouse gas emissions, projects must produce concrete positive impacts on the socio-economic level, through the creation of employment opportunities, the active involvement of the population and the strengthening of sustainable livelihoods, with particular attention to vulnerable groups, including indigenous peoples and rural communities.

The objective is to ensure that climate initiatives promote inclusion, equity and local development, while strengthening the link between environmental protection and social justice.

3.3 PERMANENCE AND ADDITIONALITY

The principles of Additionality and Permanence are fundamental pillars of the TCR Carbon Standard, designed to ensure that the carbon reductions or removals generated by afforestation and reforestation (A/R) projects are authentic, measurable and durable over time.

The principle of Additionality establishes that the climate benefits of the project must be effectively additional to a "business-as-usual" reference scenario, demonstrating that such results would not have occurred in the absence of the intervention.

Permanence, on the other hand, focuses on the need to preserve sequestered carbon over the long term. To this end, the standard provides for the adoption of preventive and corrective measures to reduce the risk of carbon reversals due to natural or anthropogenic events, such as fires, deforestation or unsustainable agricultural practices. Among these measures, the permanence buffer is also included, a technical reserve of credits intended to compensate for any losses.

3.4 TRANSPARENCY AND ACCOUNTABILITY

The principle of Transparency and Accountability is essential for consolidating trust and credibility within the Voluntary Carbon Market. This principle requires that A/R projects maintain accurate and accessible records, make the methodologies adopted publicly available, and be subjected to independent verification processes conducted by qualified third parties.

Transparency promotes the traceability of carbon credits and allows an objective assessment of the environmental effectiveness of projects. In parallel, accountability mechanisms are fundamental for addressing any critical issues such as carbon losses, fraud or operational failures, ensuring that every project operates according to high standards of integrity and compliance.

4. METHODOLOGY FOR A/R PROJECTS

4.1 PROJECT ELIGIBILITY

Prior to the implementation of an afforestation or reforestation (A/R) project, a preliminary site eligibility assessment is carried out, in order to verify that the area concerned is compatible with the project's objectives and compliant with the criteria set out in the TCR Standard.

Among the main eligibility criteria are:

- the ownership and legal availability of the land;
- the nature and condition of the soil, with particular attention to degraded, abandoned or non-forested land;
- local ecological conditions, to ensure the compatibility of activities with existing ecosystems;
- compliance with current regulations, including applicable environmental laws and territorial provisions.

This phase is fundamental to ensuring that A/R projects are feasible, sustainable and consistent with the environmental and social objectives of the standard.

4.2 REFERENCE EMISSIONS

To calculate the reference emissions, we estimate historical emissions and future emissions. For the determination of reference emissions (Baseline Emissions – BE), a quantitative estimate is made of historical and future projected emissions in the project area, assuming that no afforestation or reforestation activities are carried out ("business-as-usual" scenario).

The general formula used for the calculation of reference emissions is as follows:

BE = Historical Emissions – Projected Future Emissions

This estimate provides the reference value against which to compare the actual CO₂ removals achieved by the project, and is fundamental for the assessment of additionality and net carbon benefit.

4.3 ADDITIONALITY ASSESSMENT

The determination of additionality is an essential step in demonstrating that the climate benefits of the project would not have occurred in the absence of the A/R intervention. This is achieved through the comparison between reference emissions (BE), representing the "business-as-usual" scenario, and the actual emissions generated by the project.

The general formula for the calculation of additionality is:

Additionality = BE – Project Emissions

A positive value confirms that the project has resulted in a net reduction of emissions compared to the reference scenario, thus attesting to its environmental validity for the purposes of generating carbon credits.

4.4 CARBON ACCOUNTING

Carbon accounting represents the process through which the net carbon sequestration generated by the afforestation or reforestation (A/R) project is quantified. The calculation takes into account the main carbon flows within the forestry system, considering both biomass increments and losses.

The general formula for the calculation of Carbon Sequestration (CS) is:

CS = Total Biomass Growth – Total Biomass Decay – Harvested Wood Products

Where:

Total Biomass Growth: represents the increase in carbon stored in the above- and below-ground biomass of trees during the reference period.

Total Biomass Decay: corresponds to the loss of carbon resulting from the natural degradation of vegetation (e.g. senescence, decomposition).

Harvested Wood Products: indicates the carbon transferred to the wood products resulting from any harvesting activities conducted in the project area.

This balance makes it possible to obtain a net estimate of the carbon sequestered, useful for the generation, reporting and verification of carbon credits according to the TCR criteria.

4.5 LOSSES AND PERMANENCE

The generic formula that takes into account the reference, carbon accounting and the buffer for losses and permanence to quantify net carbon sequestration in an afforestation or reforestation (A/R) project in the context of the TCR Carbon Standard is the following:

Net Carbon Sequestration (NCS) = Total Biomass Growth – Total Biomass Decay – Harvested Wood Products – Losses – Permanence Buffer. Where:

Losses: The estimated emissions that could occur outside the project area as a result of project activities. Losses are assessed and monitored to account for emissions that could be displaced elsewhere as a result of the project.

Permanence Buffer: A reserve of carbon credits or carbon sequestration to account for potential losses due to unforeseen events, such as natural disasters or forest fires. This buffer guarantees the long-term permanence of carbon sequestration achieved by the project.

This formula takes into account the essential components of carbon sequestration in A/R projects, including biomass growth and decay, sustainable harvesting practices, the potential for emission displacement (losses) and the safeguarding of carbon conservation against unforeseen events (permanence buffer). It allows for a comprehensive assessment of the project's net carbon sequestration, considering both environmental and long-term integrity.

4.6 ABOVE- AND BELOW-GROUND BIOMASS FACTORS

The generic formula that takes into account emissions and absorption of above- and below-ground biomass for net carbon sequestration in an afforestation or reforestation (A/R) project in the TCR Carbon Standard is the following:

Net Carbon Sequestration (NCS) = (Above-Ground Biomass Growth – Above-Ground Biomass Decay) + (Below-Ground Biomass Growth – Below-Ground Biomass Decay) – Harvested Wood Products – Losses – Permanence Buffer. Where:

Above-Ground Biomass Growth: The increase in carbon stored in the above-ground part of trees through afforestation or reforestation activities.

Above-Ground Biomass Decay: The loss of carbon from the above-ground part of trees due to natural decay or decomposition.

Below-Ground Biomass Growth: The increase in carbon stored in the below-ground part of trees, including roots and soil organic matter.

Below-Ground Biomass Decay: The loss of carbon from the below-ground part of trees and soil organic matter due to natural processes.

This extended formula takes into account not only above-ground carbon, but also below-ground carbon, recognising the vital role that roots and soil play in carbon sequestration. It allows for a more comprehensive assessment of the project's net carbon sequestration, considering both above- and below-ground components, while accounting for potential emission displacements and ensuring long-term permanence.

4.7 REGIONAL CLIMATE AND POLLUTION FACTORS

Considering regional climate and pollution factors as parameters in the formula for net carbon sequestration in an afforestation or reforestation (A/R) project is a complex but valuable addition. These factors can influence a project's carbon sequestration capacity and its overall impact on the environment. The generic formula that takes these factors into account is the following:

Net Carbon Sequestration (NCS) = (Above-Ground Biomass Growth – Above-Ground Biomass Decay) + (Below-Ground Biomass Growth – Below-Ground Biomass Decay) – Harvested Wood Products – Losses – Permanence Buffer + Climate and Pollution Adjustment. Where:

Climate and Pollution Adjustment: This term represents the impact of regional climatic conditions and pollution levels on carbon sequestration. It can be calculated based on scientific models and local data. Positive values indicate an increase in carbon sequestration due to favourable climatic conditions and lower pollution levels, while negative values indicate reduced sequestration capacity due to unfavourable conditions.

To calculate the "Climate and Pollution Adjustment", it is necessary to consider factors such as temperature, precipitation, CO₂ concentration and air quality specific to the project region. The adjustment can be quantified on the basis of research and modelling relevant to the area.

By incorporating these adjustments, we take into account the local environmental context, which can significantly influence tree growth and health and, consequently, their capacity to sequester carbon. This makes carbon accounting more accurate and reflects the project's sensitivity to regional climatic and pollution conditions.

4.8 MONITORING, REPORTING AND VERIFICATION (MRV)

Effective Monitoring, Reporting and Verification (MRV) constitutes an ongoing and integral aspect of the TCR Carbon Standard for A/R projects. The MRV ensures transparent and responsible management of project activities and their impact on carbon sequestration. This cycle comprises the following key components:

Monitoring

Monitoring involves the systematic collection of data relating to project activities, carbon stocks and land use changes. It is a continuous process that provides real-time information on project performance. We implement robust monitoring systems to track and record relevant metrics.

Reporting

Reporting is the structured disclosure of the data collected and project progress. We assume the responsibility of preparing periodic reports detailing project activities, carbon sequestration, compliance with standards and any deviations from the reference scenario. The reports generated are clear, comprehensive and timely.

Verification

Verification is a critical phase for validating the accuracy and credibility of the reported data and project claims. Independent third-party verifiers assess the project against the criteria of the TCR Carbon Standard, ensuring compliance with the eligibility, additionality and environmental integrity requirements. The verified data forms the basis for the issuance of carbon credits.

The MRV cycle is iterative and is conducted at regular intervals throughout the entire project lifecycle. It provides stakeholders, including investors and the wider public, with the assurance that the project meets its environmental, social and economic objectives, contributing to climate change mitigation.

4.9 AERIAL AND SATELLITE DATA ANALYSIS WITH ML AND AI

The MRV process greatly benefits from the use of aerial and satellite technology. Here is how these technologies are leveraged to achieve effective results:

Aerial Surveys:

Remote Sensing: Aerial surveys, conducted using drones equipped with remote sensing instruments (such as LiDAR, multispot cameras and hyperspectral sensors), provide high-resolution data on forest cover, vegetation health and land use changes. These sensors capture data at various wavelengths of light, enabling detailed analysis of plant health and biomass.

Orthophotos: Aerial surveys produce orthophotos, geometrically corrected and georeferenced aerial images. These images are used to map land cover changes, monitor tree growth and assess overall forest health.

Elevation Data: Aircraft equipped with LiDAR capture elevation data, valuable for measuring canopy height and calculating above-ground biomass.

Satellite Technology:

Regular Passes: Satellites offer global coverage and the ability to acquire data at regular intervals. Satellites provide high-resolution multispectral imagery that is used to monitor A/R projects on a large scale.

Time-Series Analysis: Consistent data from satellite passes allows time-series analysis, which helps monitor changes in forest cover, identify deforestation or degradation and estimate carbon sequestration trends over time.

Climate and Environmental Data: Satellites provide climate and environmental data, such as temperature, precipitation and soil moisture, which are essential for understanding the impact of regional climatic conditions on A/R projects.

Data Analysis and Modelling:

Machine Learning and AI: Aerial and satellite data are processed using machine learning and artificial intelligence algorithms to automate the detection of land use changes, estimate biomass and assess project compliance with MRV standards.

Carbon Modelling: Carbon models are calibrated using remote sensing data to more accurately estimate carbon sequestration and losses within A/R projects.

Transparency and Reporting:

Data Visualisation: Aerial and satellite data are transformed into interactive maps and visualisations, making it easier for stakeholders, including project developers and the public, to understand project progress and carbon sequestration.

Verification:

Independent Verification: Independent third-party verifiers can use the same aerial and satellite data to assess project compliance with MRV requirements. The consistency and objectivity of remotely sensed data enhance the credibility of the verification process.

By integrating aerial and satellite technology into the MRV process, A/R projects can achieve efficient, scalable and data-rich monitoring and reporting, crucial for maintaining transparency, accountability and environmental integrity in the TCR Carbon Standard.

These technologies also help to address the challenges related to large project areas and remote locations. Thanks to the ability of drones and satellites to cover vast stretches of territory, it is possible to monitor and collect data on extended areas efficiently and promptly, even in places that are difficult to reach. This makes it possible to obtain a more complete and accurate view of project activities, reducing reliance on costly ground surveys and the logistical difficulties associated with access to remote locations. Furthermore, automation and advanced data analysis through artificial intelligence and machine learning technologies make it possible to process large amounts of information quickly and efficiently, providing an in-depth analysis of environmental conditions and project performance on a large scale. These tools therefore play a fundamental role in overcoming the challenges related to the size and geographical location of afforestation or reforestation projects, thus contributing to improving the effectiveness and efficiency of climate change mitigation initiatives.

5. PROJECT ELIGIBILITY AND REQUIREMENTS

In line with the TCR Carbon Standard Methodology, projects aiming to obtain the issuance of carbon credits must adhere to rigorous eligibility criteria. The first step in this assessment process involves the submission of project information and documentation through Carborea's Onboarding Form, accessible through the proprietary technology platform. All projects are required to provide complete information, as described below:

5.1 PROJECT INFORMATION AND APPLICANT DETAILS

To ensure transparency and facilitate efficient communication, projects must provide the following information:

Applicant Information and KYC/KYB Process

- Essential applicant details, including name, email address, identification number and physical address, whether the applicant represents a legal entity or is an individual. These details serve as the basis for communication throughout the entire assessment process.

- Proof of identity for the applicant, comprising scanned copies of relevant identification documents. For individuals, this may involve the submission of a scanned copy of a local identity document, passport or driving licence. Legal entities should provide proof of incorporation or official registry extracts.

Project Representative Information

- Identification information for the project representative, who will sign agreements on behalf of the project owner. In scenarios where the project owner is a legal entity, the representative may be any individual with the legal capacity to act on behalf of the entity. In cases where the project owner is an individual, they are considered the representative and will sign the agreements accordingly.
- Proof of identity for the representative, with acceptable proofs including a scanned copy of their local identity document, passport or driving licence.

Project Details

- A complete description of the project outlining the type of commercial or project activities undertaken. This description will serve Carborea in assessing whether the activities are in line with the principle of Additionality.
- Detailed project specifications, including the total size of the land designated for onboarding under the current TCR, the precise coordinates of the mapped land using the methods recommended by Carborea as indicated in the Basic Guidelines for Project Mapping, the physical address of the land and specific data relating to the number, species and age of the trees or plants engaged in CO₂ absorption through Additional Activities.

Proof of Long-Term Land Rights

- Projects are required to provide verifiable evidence of long-term land rights. Such rights may include legal ownership, lease contracts or similar legal agreements with a duration of at least 20 years from the date of project onboarding.
- Land rights verification documents, which may include a scanned copy of the agreement, notarial deed or public registry information.

Proof of Tree Purchase

- In cases where Additional Activities involve tree planting, projects are required to submit documented proof of tree purchase.
- Suitable proofs for tree purchase, involving the planting of trees or other Additional Activities, may include receipts or purchase slips.

Images and/or Video

- A visual component of project documentation requires the submission of at least two images or videos per species of plant or tree involved in the Additional Activities.

Long-Term Commitment

- Projects must explicitly specify the number of years during which they intend to maintain green assets and carbon offsetting. This commitment should extend for a minimum of 20 years from the date of project onboarding.

Emissions Assessment

- In accordance with Step 4.5 of the descriptive section of the Methodology, projects are called upon to identify and assess the possible emissions associated with the project.

5.2 PROJECT CHARACTERISTICS

For a project to be considered for onboarding, it must meet a series of specific requirements as detailed in the current document. These requirements apply universally to all projects seeking the issuance of credits:

Additionality

Projects are required to provide convincing evidence that the activities aimed at improved CO₂ offsetting would not have occurred in the absence of the Additional Activities designated for onboarding and tokenisation. A comprehensive understanding of Additionality is provided in the descriptive section of the TCR Carbon Standard Methodology. Due to the abstract nature of Additionality, Carborea will assess on a case-by-case basis, considering the unique circumstances of each project to determine compliance with this requirement. Projects unable to demonstrate Additionality will not be considered for onboarding.

Long-Term Carbon Offsetting

Carbon offsetting activities must be maintained and diligently upheld for a minimum duration of 20 years, starting from the project onboarding date. As previously emphasised in the TCR, this requirement means that Carborea will consider exclusively projects with a commitment to maintaining offsetting activities for the established 20-year period. Projects with shorter durations will not be considered for onboarding.

Absence of Intentional Deforestation

The project site must not have undergone deliberate deforestation in the five years preceding the start of the project or the commencement of Additional Activities. Carborea will carry out a verification using historical satellite data to confirm compliance. It is important to note that only intentional deforestation activities carried out by the project owner constitute grounds for ineligibility of a project. Incidents involving accidents or natural disasters are not considered deterrents when assessing the eligibility of a project for onboarding.

Use of Native Vegetation

Tree planting projects must employ exclusively vegetation types native to the project area.

Non-Invasive Species

Agroforestry and orchard projects may use exclusively non-invasive species. Although exotic species are permitted, they must also be non-invasive. Native species are preferred when possible. Projects that do not comply with these specifications will not be considered for onboarding.

Absence of Previous Carbon Offsetting Units

Projects are excluded from the creation or receipt of any form of carbon offsetting units, including carbon offsets, carbon credits or carbon tokens, in exchange for the Additional Activities designated for tokenisation. This prohibition extends to carbon offsetting units obtained from other institutions such as Verra or Gold Standard. The objective is to avoid double counting of carbon emission reductions. Carborea will verify compliance with this requirement through various registries. It is important to emphasise that this provision does not prevent projects from interacting with these institutions for purposes unrelated to CO₂ offsetting.

These rigorous eligibility criteria and project characteristics are designed to ensure that projects are aligned with the principles underlying Carborea's TCR Standard. By promoting transparency, environmental sustainability and the long-term effectiveness of carbon offsetting activities, the Standard facilitates the assessment of each project's compliance with these requirements. This approach ensures that only projects that meet the Standard's rigorous criteria are considered for the issuance of carbon credits.

5.3 PROJECT DAMAGE AND RISK MITIGATION

In strict compliance with the guidelines outlined in the Onboarding Agreement, onboarding projects are entrusted with the responsibility of implementing proactive measures to safeguard the integrity of their carbon offsetting activities. In particular, projects are required to carry out the planting of an additional 10% of each plant species or to allocate 10% of each plant species without tokenisation. This proactive approach serves as an essential contingency plan to address the potential

occurrence of plant damage, ensuring that Tokens remain firmly associated with tangible trees and their CO₂ absorption.

Contingency Measures

In the unfortunate scenario where plant damage exceeds the reserves initially set aside, Carborea firmly commits to working closely with the project owner. This collaboration is based on good faith and seeks to identify the most appropriate solution that aligns with the interests of both parties.

This collaborative approach underlines the fundamental importance of transparent and open communication between Carborea and project owners. It reinforces the idea that project success is a shared commitment and that, by working together, challenges and setbacks can be effectively addressed.

"Black-Listing" of Tokens

In cases where reaching a mutually acceptable resolution proves difficult, Carborea maintains the authority to employ a "black-listing" mechanism for the affected Tokens. This mechanism involves marking the Tokens in such a way that they become non-functional, and the modification of their status is made evident to the public.

The purpose of this measure is to safeguard the integrity and credibility of the carbon offsetting ecosystem. By marking Tokens that are no longer associated with valid trees and CO₂ absorption, Carborea ensures that such Tokens do not continue to circulate within the market. This proactive step is fundamental to preventing the trading or inadvertent use of Tokens that no longer represent true carbon offsetting assets.

Project Responsibility

Project owners must assume full responsibility for any cases of plant damage or black-listed Tokens that may arise in their interactions with third-party buyers. The decision to initiate Token sales remains entirely within the competence of project owners, who assume all associated risks.

Carborea is unable to provide compensation to either project owners or potential third-party buyers in cases involving damaged or black-listed Tokens. This limitation stems from Carborea's commitment to offering onboarding services at no cost and from the lack of effective and material control over potential plant damage.

Therefore, it is the responsibility of both projects and potential buyers to exercise caution and diligence in all Token-related transactions. This collaborative and transparent approach ensures that Carborea's TCR Standard continues to maintain its solidity and reliability within the Voluntary Carbon Market, even in the face of unforeseen challenges.

6. THE ROLE OF BLOCKCHAIN TECHNOLOGY

Ensuring the accuracy and reliability of carbon credit issuance is fundamental, and blockchain technology plays a crucial role in this process:

6.1 METHODOLOGY APPROVAL

Smart Contracts for Verification: Smart Contracts developed on blockchain are used for the automated verification of project compliance with Carborea's TCR Standard. This ensures that carbon credits are issued only when all criteria are met.

Immutable Records: The blockchain creates an immutable record of methodology approval, making it transparent and resistant to manipulation.

6.2 MONITORING AND VERIFICATION

Real-Time Data: IoT sensors integrated with blockchain technology provide real-time data on project activities, which can be used for continuous verification.

Blockchain Auditing: Third-party verifiers can audit the blockchain registry to verify the issuance and transfer of carbon credits, increasing trust among stakeholders.

6.3 ISSUANCE OF CARBON CREDITS

Smart Contract-Based Issuance: Carbon credits are issued automatically through smart contracts once the project's carbon sequestration has been independently verified. Each credit represents one tonne of CO₂eq.

Public Registry: The issuance of carbon credits should be recorded on a public blockchain registry, allowing stakeholders to track the creation and ownership of credits.

6.4 RETIREMENT OF CARBON CREDITS

Blockchain-Notarised Retirement: The blockchain facilitates the retirement of carbon credits at the time of use, creating a verifiable and tamper-proof record of retired credits.

Interoperability: The monitoring of blockchain transactions across different chains has made possible the interoperability of carbon credit retirement systems with various market platforms to ensure uniform credit retirement.

6.5 TRANSPARENCY AND PUBLIC REGISTRY

Public Registry: Maintaining a public blockchain registry in which all issued and retired carbon credits are recorded, providing a transparent and easily accessible record of carbon credit activity.

Ownership Transparency: The blockchain allows stakeholders to trace the ownership history of carbon credits, promoting transparency and reducing the risk of double counting.

7. MARKET AND STAKEHOLDER ENGAGEMENT

To ensure the success and sustainability of the carbon credit market and the surrounding ecosystem, it is essential to actively engage the various actors involved:

Stakeholder Participation: Actively engaging a wide range of stakeholders, including government organisations, businesses, non-governmental organisations (NGOs), investors and local communities. This engagement makes it possible to obtain a diversified perspective and to maximise adherence to and support for the carbon credit market.

Education and Awareness: Providing clear and accessible information on the benefits and functioning of the carbon credit market. Promoting awareness of the importance of reducing carbon emissions and purchasing carbon credits as a tool for mitigating climate change.

Incentives and Support: Offering incentives and support to interested actors to actively participate in the carbon credit market. These incentives may include tax incentives, concessional financing, public recognition and training and technical assistance programmes.

Transparency and Communication: Maintaining open and transparent communication on all activities and decisions of the carbon credit market. Providing regular reports on market performance and results achieved, as well as on evolving policies and regulations.

Local Community Engagement: Actively involving local communities in decisions and activities related to the carbon credit market, especially if projects are located in their areas. Ensuring that local communities derive tangible benefits from the projects, such as employment opportunities, local economic development and environmental improvements.

International Collaboration: Promoting international collaboration and cooperation in the carbon credit market, in order to maximise the effectiveness of climate change mitigation actions at the global level. Facilitating the exchange of knowledge, experiences and resources between countries and regions.

The active involvement and support of a wide range of stakeholders are fundamental for the long-term growth and success of the carbon credit market, making it possible to address complex challenges and maximise positive impacts on the climate and environment.

8. GOVERNANCE AND ADMINISTRATION

8.1 GOVERNING BODY

Carborea's governing body is responsible for the strategic management and continuous improvement of the standard. It is composed of:

Steering Committee: composed of experts in sustainability, blockchain, carbon markets and environmental sciences.

Technical Assessment Group (TAG): analyses new methodologies and verifies compliance with additionality and permanence criteria.

Transparency and Compliance Council: ensures that the TCR Carbon Standard operates according to principles of fairness and transparency, assessing reports and complaints.

8.2 STANDARDS SECRETARIAT

The Standards Secretariat is an operational entity that manages the day-to-day activities of the standard, including:

- Review and integration of the standard based on regulatory and technological developments.
- Management of the project and carbon credit registry.

- Coordination with interested parties, including project developers, third-party verifiers and investors.

8.3 REVIEW AND REASSESSMENT PROCESS

The TCR is subject to periodic reviews to ensure continued alignment with scientific and regulatory developments.

Periodic Review: integration of improvements based on new scientific evidence and stakeholder feedback.

Triennial Audit: independent assessment of the impact of the standard and the methodologies adopted.

Public Consultations: stakeholders can propose changes or improvements through a structured feedback process.

8.4 CONFLICT RESOLUTION

To manage disputes between project developers, credit buyers and verifiers, the TCR Carbon Standard adopts a dispute resolution system that includes:

Mediation: attempt at internal resolution between the parties involved.

Arbitration: in the event of failure to reach an agreement, an independent committee assesses the case and proposes a solution.

Exclusion from the Registry: for projects or actors that violate the standards or fail to comply with the conformity criteria.

8.5 APPEALS MECHANISM

Project developers or other stakeholders who do not agree with a decision of the TCR Carbon Standard may submit an appeal. The appeals process provides for:

- Submission of a formal request with documented evidence.
- Review of the request by the Transparency and Compliance Council.
- Final binding decision within 90 days.

9. COMPLIANCE WITH EXISTING STANDARDS

9.1 HARMONISATION EFFORTS

The TCR is designed to be interoperable with the main standards of the voluntary carbon market, ensuring the maximum integrity of the credits issued. The standard aligns with:

Verra VCS (Voluntary Carbon Standard): adopts similar additionality, permanence and verifiability criteria.

Gold Standard: compatibility with credit certification methodologies.

ISO 14064-2: adherence to international principles for monitoring and verifying CO₂ reductions.

CDM (Clean Development Mechanism) of the UNFCCC: compatibility with emission accounting mechanisms.

10. CONCLUSIONS

The TCR Standard, adapted for the Voluntary Carbon Market, underlines its alignment with climate action by placing a central focus on Additionality as a cardinal principle.

It operates in a dynamic context, leveraging a wide range of references, from global standards to regional regulations, promoting a transparent and interdisciplinary approach.

This standard is distinguished by its commitment to risk mitigation, with innovative mechanisms for addressing project damage and safeguarding the credibility of tokens.

It emphasises community consensus and environmental restoration, enabling projects to contribute actively. The attached disclaimer reinforces individual responsibility and underlines the need for independent research.

Finally, Carborea's long-term commitment to sustainability, transparency and collaboration elevates this methodology to a valuable resource for stakeholders guiding climate change mitigation in the Voluntary Carbon Market.

REFERENCES

- UNI EN ISO 14064-1:2019 Greenhouse gases – Part 1: Specification with guidance at the organisation level for quantification and reporting of greenhouse gas emissions and removals.
- UNI EN ISO 14064-2:2019 Greenhouse gases – Part 2: Specification with guidance at the project level for quantification, monitoring and reporting of greenhouse gas emission reductions or removal enhancements.
- UNI EN ISO 14065:2013 Greenhouse gases – Requirements for greenhouse gas validation and verification bodies for use in accreditation or other forms of recognition.
- UNFCCC (2011) Methodological tool. Tool for the demonstration and assessment of additionality (Version 06.0.0) CDM Executive Board.
- References for the "Organic Agriculture" methodology:
 - IPCC 2006 Guidelines for National Greenhouse Gas Inventories
 - IPCC Good Practice Guidance for Land Use, Land-Use Change and Forestry
 - FSC (Forest Stewardship Council) Standard for Good Forest Management in Italy
 - VCS v3.2 Standard (Voluntary Carbon Standard)
 - VCS Requirements for Agriculture, Forestry and Other Land Uses (AFOLU)
 - VCS AFOLU Non-Permanence Risk Tool v4.2 – Analysis and Buffer Determination
 - UK Woodland Carbon Code
 - Council Regulation (EC) No 834/2007 of 28 June 2007 on organic production
 - Commission Regulation (EC) No 889/2008 of 5 September 2008 laying down detailed rules for the implementation of Council Regulation (EC) No 834/2007
 - Carbon Sequestration Model in Marsh Waters of the Venetian Lagoon, Italy (2013). Doimi M. et al. Journal of Environmental Science and Engineering B2
 - ISMEA; MIPAFF (2013); Survey on studies and research on the mitigation potential of certain agricultural practices and operations
 - Veneto Region, Veneto Agriculture (2010). Crop rotations, associations and soil fertility in organic farming
 - Regulation 710/09/EC Organic Aquaculture